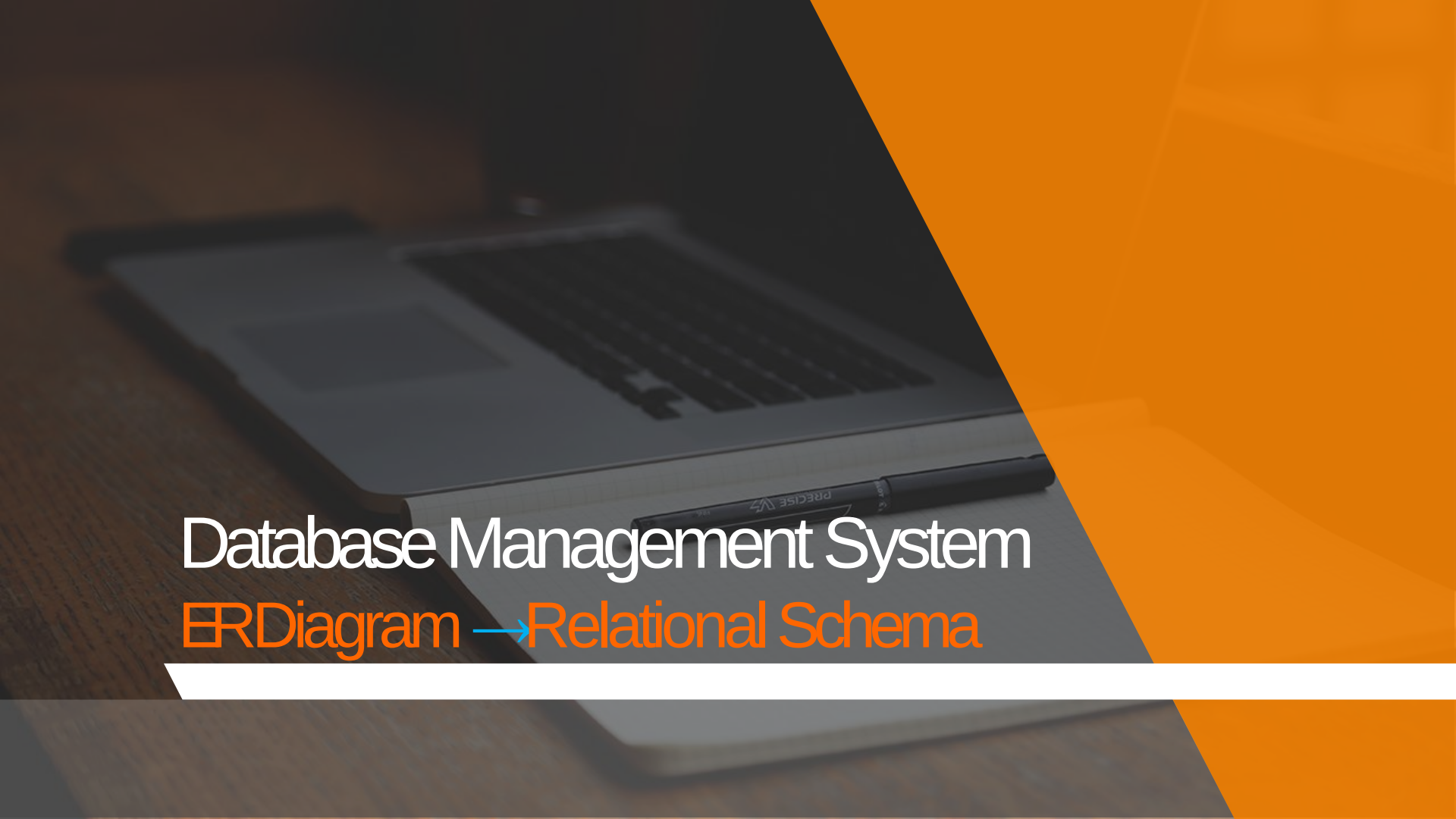
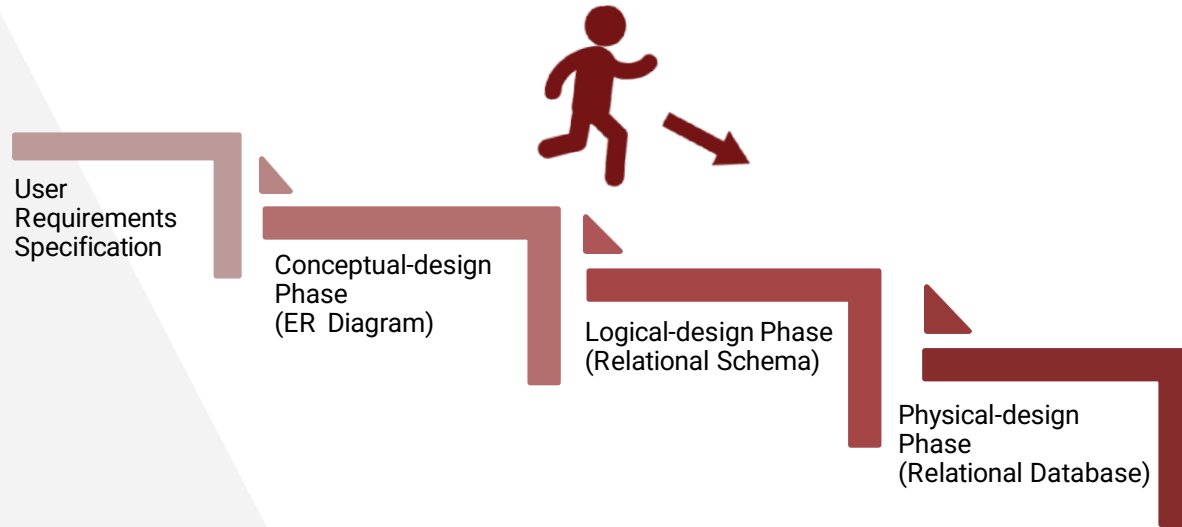




Database Management System

ERDiagram → Relational Schema

Design Phases



Database Schema & Data Model

Database Schema

Describes the overall design of the database at different levels(physical, logical, view)

Data Model

Describes a way to design database at physical, logical and view level. It is a collection of conceptual tools and languages for describing data, data relationships, data semantics, and consistency constraints.

High-level conceptual data model ⑦ ER Model

Record based logical data model ⑦ Relational Model

E-R Diagram **vs** Relational Schema

ER Diagram

The main components of ER Model are:

- Entity Set – set of similar entities
- Relationship Set – set of similar relationships
- Attributes – including different types of keys

ER Model describes Mapping Cardinality.

Relational Schema

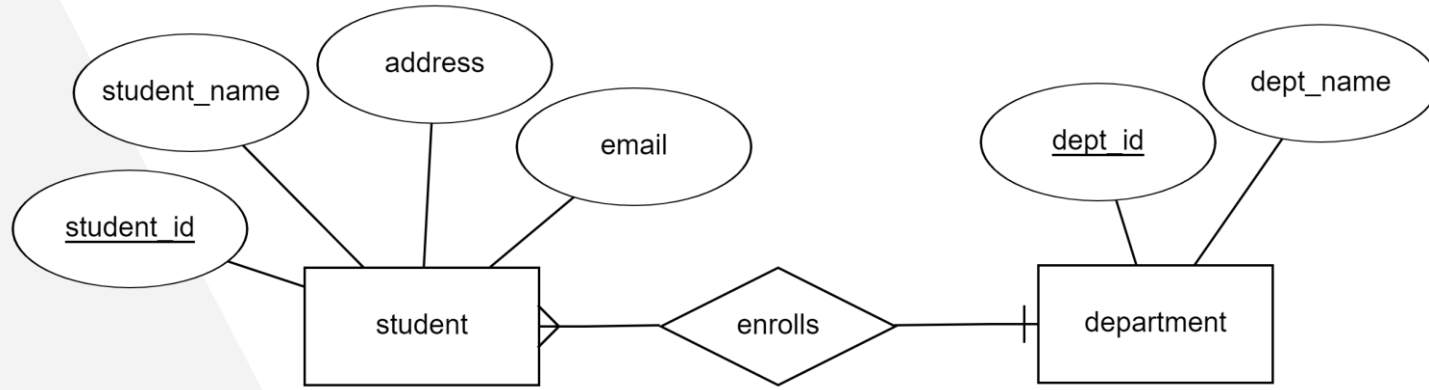
The main components of Relational Model are:

- Relations (tables)
- Tuple (row, unordered)
- Attributes (columns, unordered)
- Domain (column type)
- Primary Key
- Foreign Key

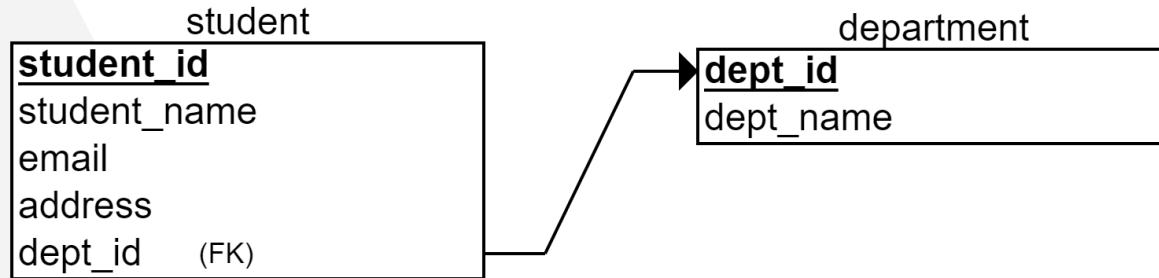
Relational Model does not describe Mapping Cardinality.

E-R Diagram vs Relational Schema

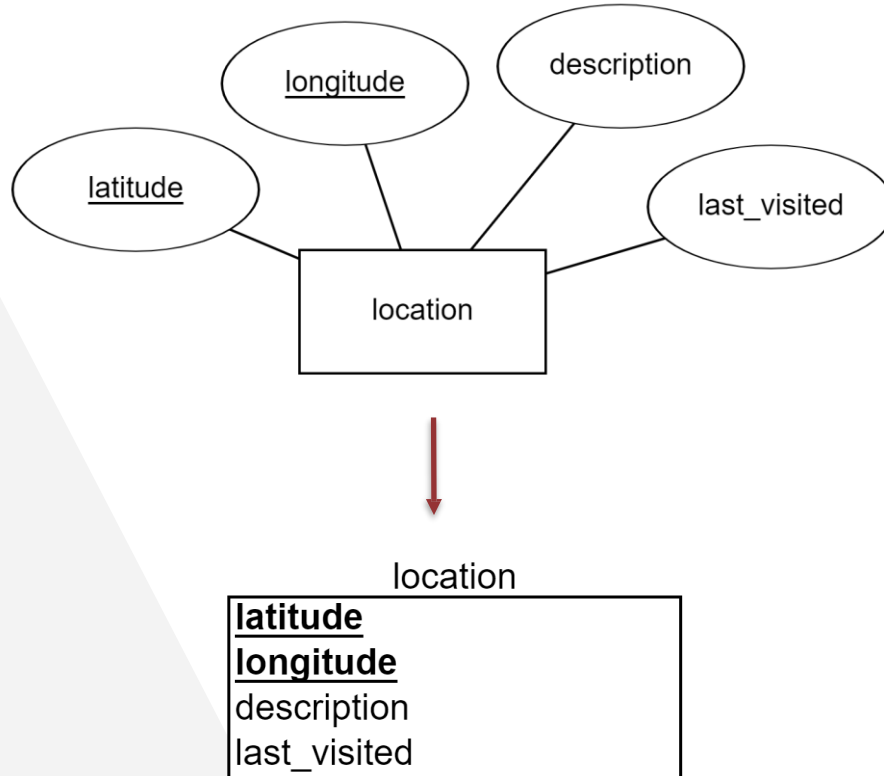
ER Diagram :



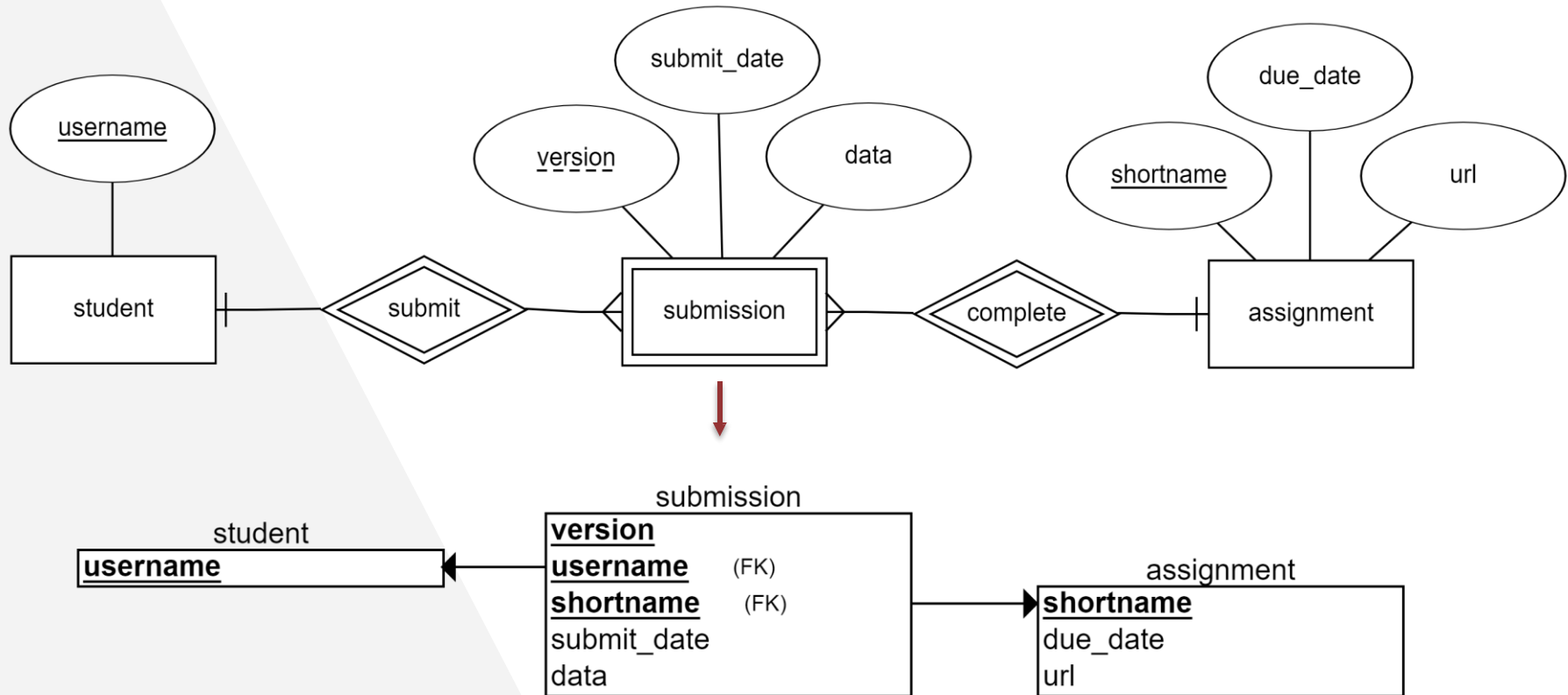
Relational Schema :



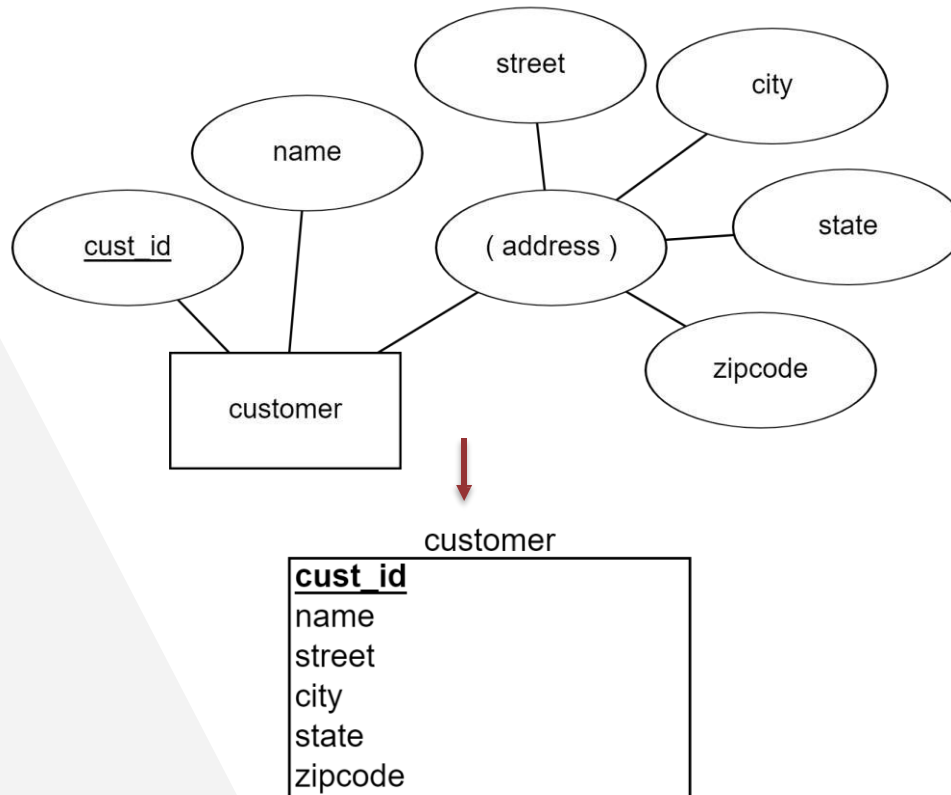
Relational Schema >> Strong Entity Set



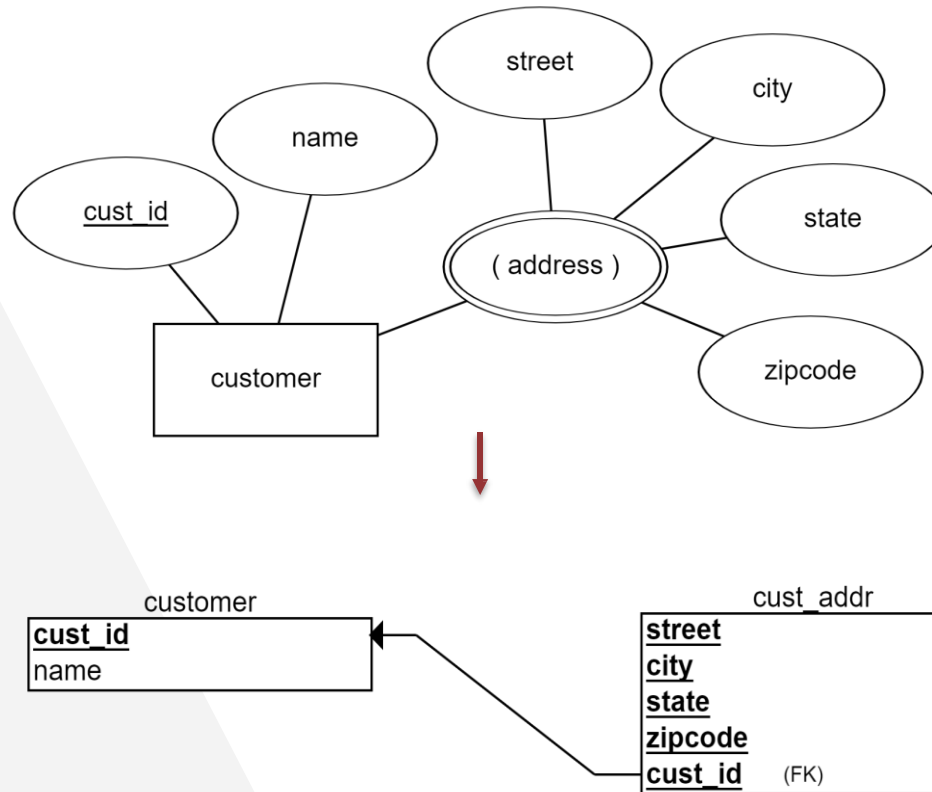
Relational Schema >> Weak Entity Set



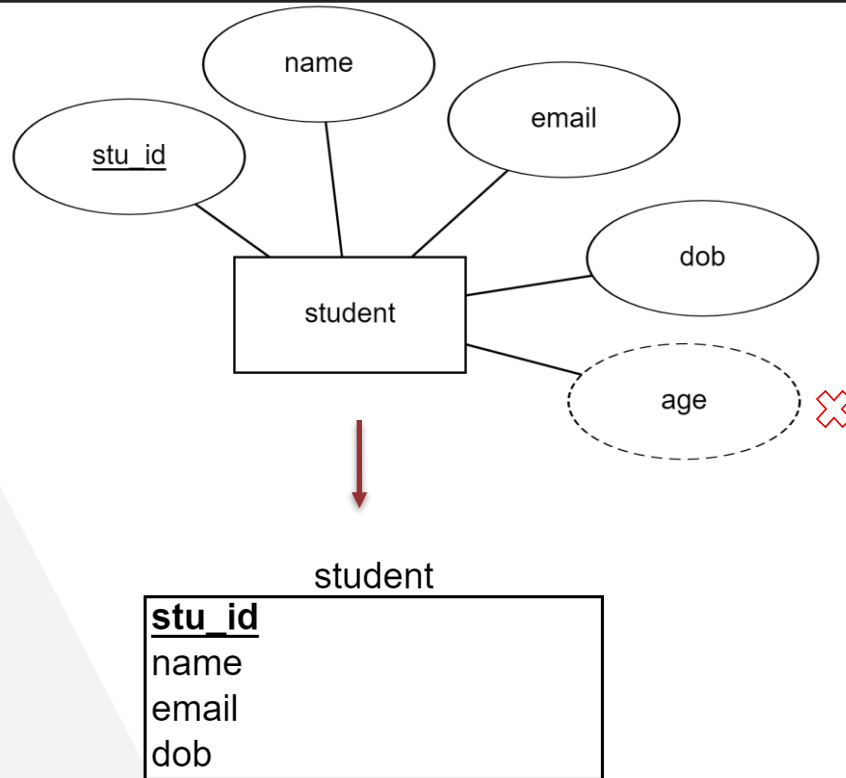
Relational Schema >> Composite Attribute



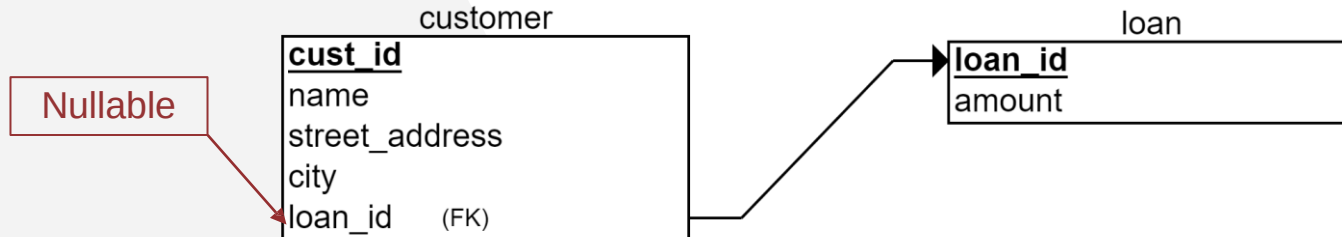
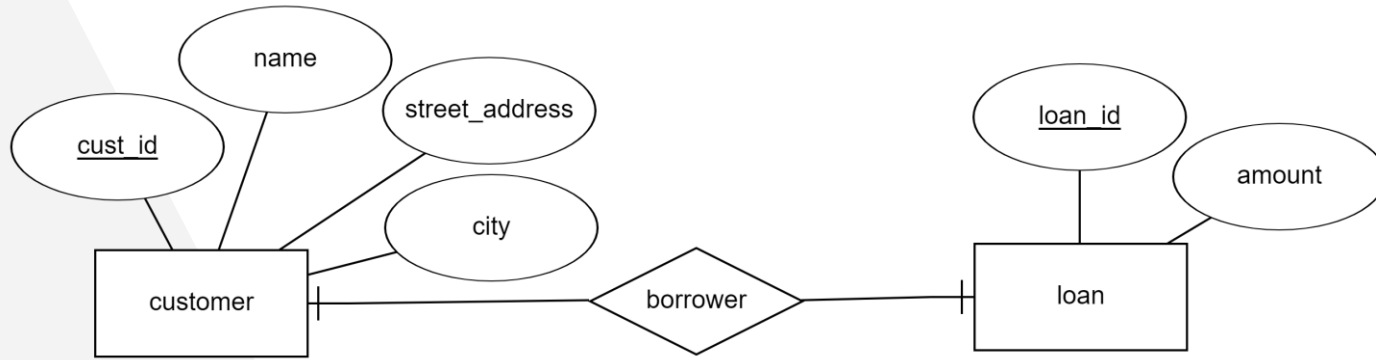
Relational Schema >> Multivalued Attribute



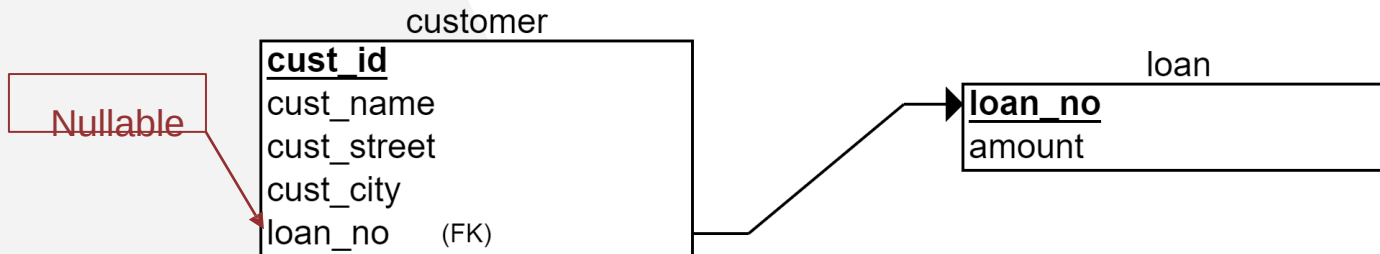
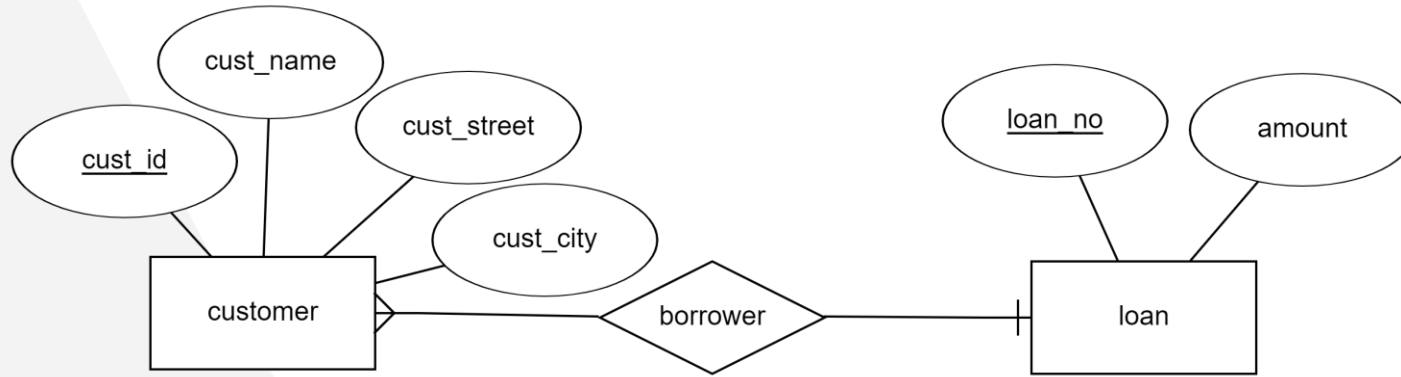
Relational Schema >> Derived Attribute



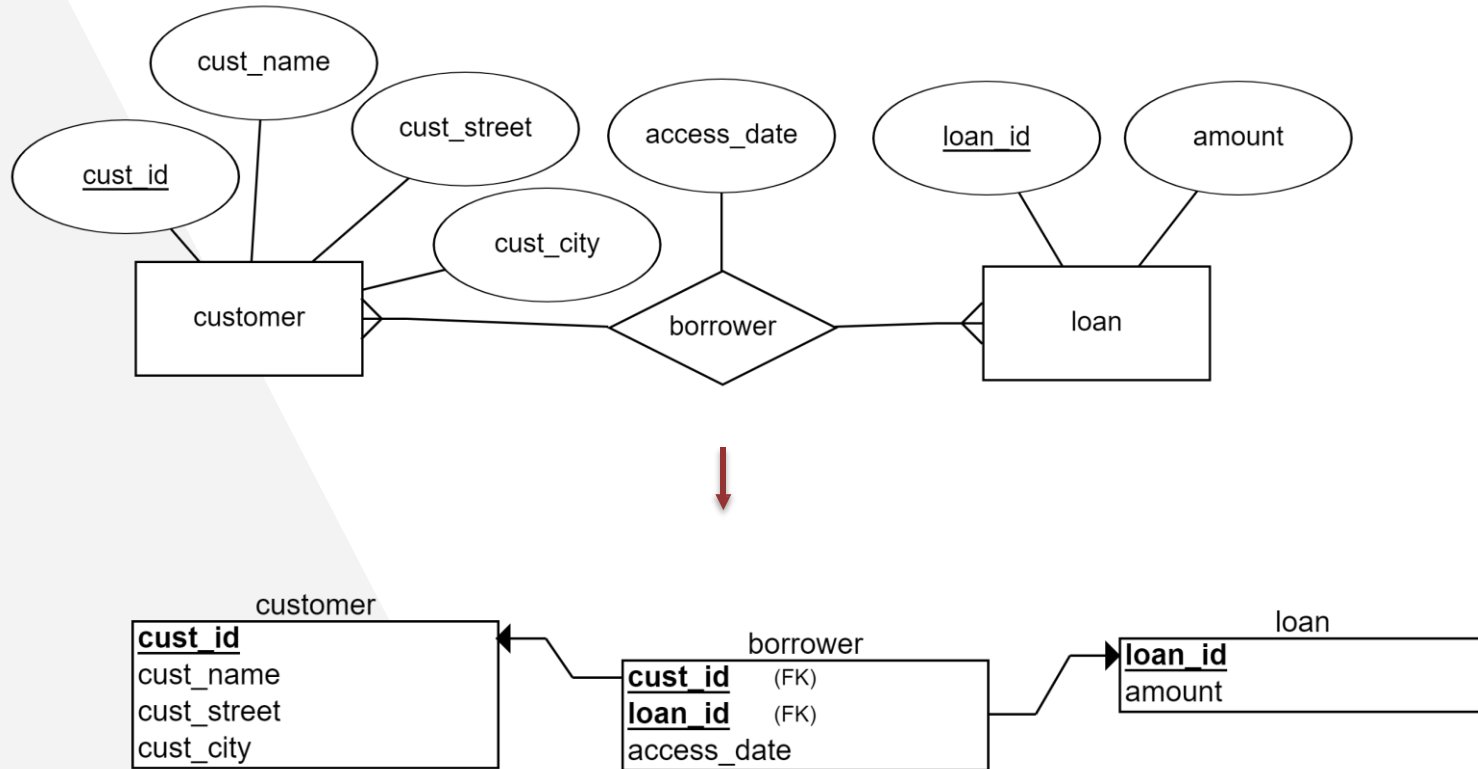
Relational Schema >> One to One Relationship Set



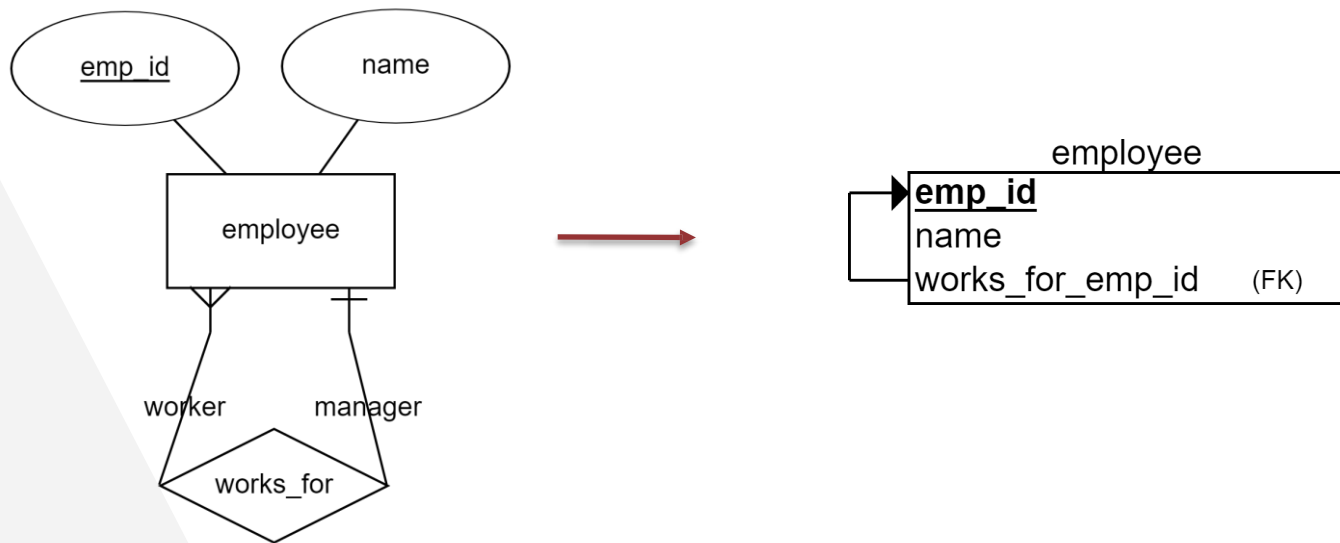
Relational Schema >> One to Many Relationship Set



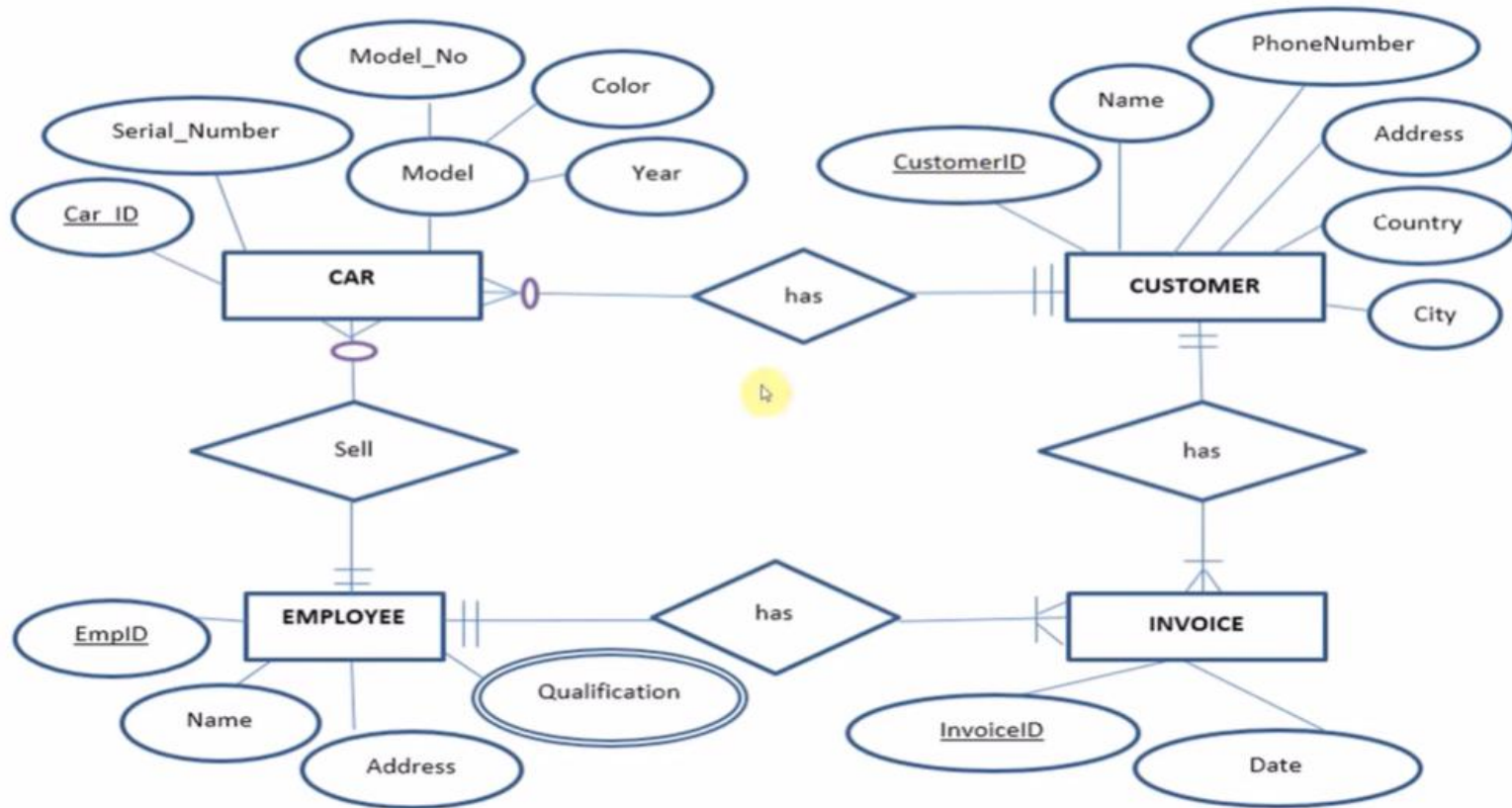
Relational Schema >> Many to Many Relationship Set



Relational Schema >> Recursive Relationship Set

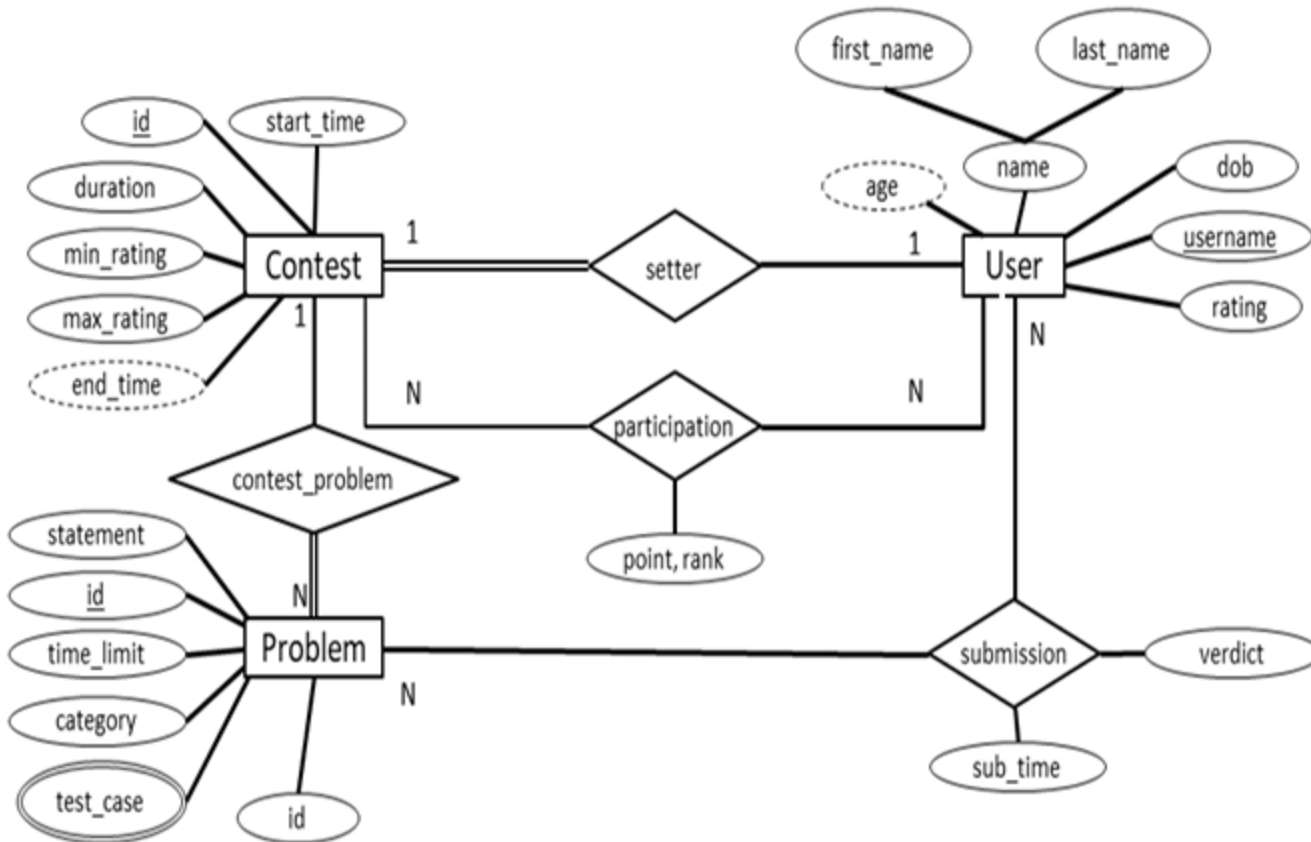


Transform ERD to Relational Schema

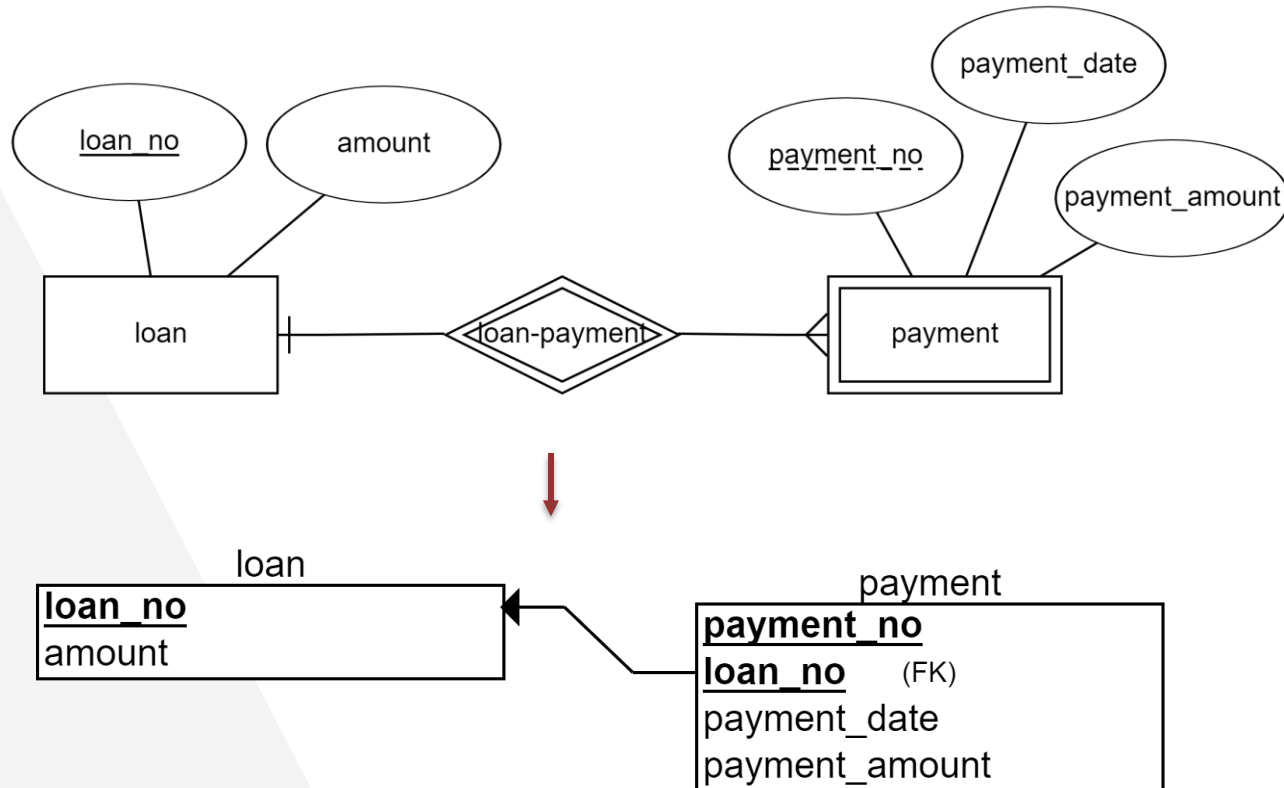


2. **Draw** the corresponding schema diagram based on the erd given in Figure 1.

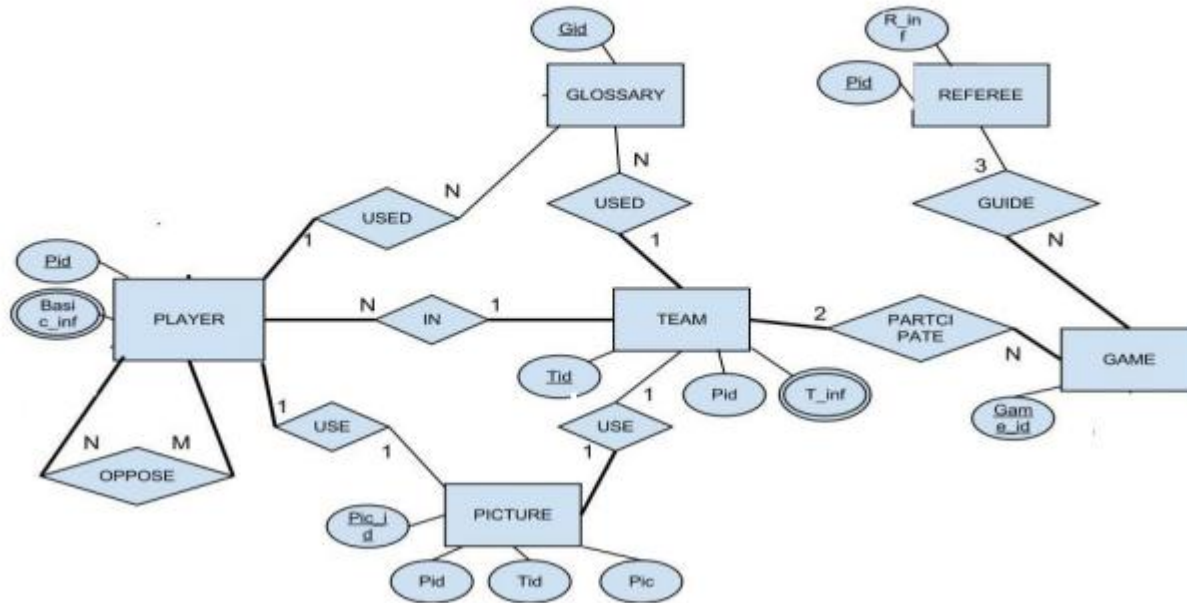
5



Relational Schema >> Identifying Relationship Set



2. Draw the corresponding schema diagram for the ERD below,



E-R Diagram → Relational Schema

